Comparative Analysis of Machine Learning Techniques for Credit Score Prediction

In this coursework, you will explore the application of machine learning techniques within the domain of financial risk assessment. Your primary objective is to conduct a rigorous comparative study evaluating how different algorithms perform when predicting consumer creditworthiness across two distinct data environments.

Task Description

1. Dataset Acquisition & Selection: You will work with two separate data sources for this assignment:

i) **Provided Dataset:** You must utilize the credit scoring dataset provided in google classroom, which contains relevant credit-related features and corresponding creditworthiness labels.

ii) **External (chosen) Dataset**: You are required to independently find and download an additional, appropriate credit scoring dataset from reputable open repositories (such as Kaggle, the UCI Machine Learning Repository, or OpenML).

2. Methodology & Implementation: You will apply machine learning techniques to both datasets. This involves performing necessary data preprocessing and feature engineering, followed by training a variety of machine learning algorithms (e.g., Logistic Regression, Decision Trees, Ensemble Methods, or Neural Networks).

3. Evaluation & Comparative Analysis: You must evaluate and compare your results across both datasets. Analyze the effectiveness of your implemented models using robust performance metrics (such as Accuracy, F1-Score, Precision, Recall, and ROC-AUC). Your task is to perform controlled experiments and critically analyze how model performance, stability, and feature importance vary between the provided data and your independently sourced dataset.

4. Conclusion: Draw evidence-based conclusions regarding the generalization capabilities of different machine learning approaches, highlighting which algorithms prove most robust across different credit scoring data environments.

Guidelines

i. Familiarise yourself with the concept of credit scoring and its significance in the financial industry. Explore and summarize the application of machine learning techniques for credit scoring from different credible sources, mainly conference and journal papers.

ii. Explore the given dataset and understand its structure and content. Identify the credit-related features available and their potential relevance in predicting creditworthiness.

iii. Pre-process the provided dataset (and also second chosen dataset) by addressing any missing values, outliers, or inconsistencies (if necessary). Apply appropriate techniques such as data cleaning, normalization, and feature scaling to ensure the dataset is suitable for machine learning algorithms.

iv. Select at least four different machine learning algorithms suitable for credit scoring, considering their ability to handle classification tasks and their relevance to credit scoring. Justify your choice

of algorithms based on their strengths and weaknesses. You need to apply at least two algorithms covered in the lecture.

v. Split the dataset, ensuring an appropriate distribution of positive and negative samples in the respective sets. Apply the selected machine learning algorithms to the training set and evaluate their performance on the validation and testing set using suitable evaluation metrics (e.g., accuracy, precision, recall, F1 score). You may choose to compare this approach using cross-validation with k-folds.

vi. Compare and analyse the results obtained from the different algorithms. Discuss which algorithm(s) performed better in credit scoring based on the evaluation metrics and provide reasons for their performance. Consider factors such as model complexity, bias-variance trade-off, and interpretability.

vii. In your experiments, you need to investigate the impact of hyperparameter tuning on the performance of the machine learning models.

viii. Based on your findings and analysis, provide recommendations for financial institutions regarding the application and adoption of machine learning in credit scoring. Consider the benefits and drawbacks of using machine learning models in this context and suggest ways to overcome challenges and limitations.

ix. Conclude your coursework by summarizing the key insights gained from your experiments and discussing the potential future developments and advancements in machine learning for credit scoring.

x. Detailed explanations, justifications, and analysis must be supported by references to research papers (mainly), books, or other credible sources.

Submission

Submission deadline will be defined in Google Classroom.

You need to submit 1) a report and 2) a video/class presentation, as per below:

1. **Report:** You need to submit a report, with the following main headings
    - Introduction, Literature Review, Methodology, Experiments, Results and Discussion, Recommendations, Conclusion, References.

2. Jupyter Notebook: Upload along with your report.

3. ***Video/class Presentation:** You need to prepare a short presentation of **maximum 6 minutes**, which briefly describes your approaches and focuses mainly on the results achieved by the models you implemented for classification and regression problems. The video must show slides and the webcam view (face) of the student describing/explaining the work. You may use the following software for recording your presentation: https://screencast-o-matic.com/

NOTES:

- You need to submit (upload) your report, code(notebook/.ipynb) recorded video presentation in google classroom only. Do not compress (zip/rar) your report, notebook and the video.

- *The recorded video presentation may be replaced by a class presentation. Exact details will be provided in due time.
- This is a groupwork, with **maximum two students** per group.
- Use of Generative AI must be clearly specified (e.g. which sections, paragraphs or other parts or the report as well as for code generation/debugging). Write a clear statement at the start of the report in case you make use of GenAI tools.

Important Notice

Data Usage Statement: **The dataset provided for this coursework is strictly intended for academic purposes and should only be used within the context of this coursework for the module "Supervised Machine Learning Techniques" offered at the UTM for Semester 1 – 2026 (Semester period APR 2026 to AUG 2026). The dataset is not to be reproduced, shared, or used for any other purposes, commercial or non-commercial, without explicit permission from the dataset owner or source. Any unauthorized use of the dataset beyond the scope of this coursework is strictly prohibited.**

Marking Scheme (100 marks - 60%)

- Literature review: 10
- Methodology
    - Data Pre-processing/Manipulation: 10
    - Choice of Algorithms: 10
    - Experimental Design: 10
- Experiments, Results & Evaluation: 40
- Recommendations (includes challenges, limitations & future works): 20

*The weight for this coursework is subject to change.